

Dr Tom D. Pering, *SFHEA*

Links: [ORCID](#), **Personal Website -** [VolcanoBlog](#), **Group Website -** [VolcanoTech](#)

Overview:

My high-impact research, which was submitted to REF2021 in the UK, revolves around the remote sensing of sulphur dioxide using low-cost ultraviolet (UV) cameras and modelling magma-gas behaviour. I lead the Sheffield volcano group and company “VolcanoTech” who are currently installing the next generation of monitoring equipment worldwide to assist in eruption forecasting.

Academic Employment and Education:

CEO & Founder VolcanoTech Limited , Sheffield, UK	01/2026
Senior Lecturer , University of Sheffield	01/2023-
Lecturer , <i>University of Sheffield</i>	2018-2022
Teaching Associate , <i>University of Sheffield</i>	2016-2018
PhD Volcanology awarded April 2016 , Bubbles in Basalts, <i>University of Sheffield</i>	2012-2015
MSc Volcanology and Geological Hazards (Distinction) , <i>Lancaster University</i>	2010-2011
BSc (Hons) Physical Geography (2:1) , <i>University of Reading</i>	2007-2010

Funding & Income:

PI, 2026-ongoing: M4B	£40k
PI, 2026-ongoing: EPSRC – Impact	£53k
PI, 2026-ongoing: ESRC – Impact	£5.5k
PI, 2024-2025: NTT Management grant for VolcanoTech	£35k
PI, 2025-ongoing: Consultancy – Gas sampling for Milan	£2k
PI, 2025-ongoing: Consultancy – Gas sampling at Santorini volcano	£2.2k
Co-I, 2024-2025: ISPF Institutional Support Grant (ODA)	£45k
PI, 2024-2025: EPSRC KE – Automated processing of UV Imagery	£52k
Supervisor (2nd), 2023-2027: EPSRC – Machine learning of UV Imagery	£56k + fees
Co-I, 2023-2025: NSF, Multi-camera approach for monitoring volcanoes	£382k
Supervisor (1st), 2023-2027: EPSRC – Spectroscopic measurements of gases	£56k + fees
PI, 2023-2025: Consultancy – UV Cameras for LMV (France)	£4k
PI, 2023-2025: Consultancy – UV Cameras for Colima (Mexico)	£4k
PI, 2023-2025: Consultancy – UV Cameras for Columbia Uni (USA)	£2k+
PI, 2023-2025: Royal Society – Enhancing monitoring at Merapi (Indonesia)	£12k
PI, 2023-2024: EPSRC KE – Enhancing the capabilities of UV cameras at Kilauea	£55k
PI, 2021-2022: Consultancy – UV Cameras for Hawaii (USA)	£8k
PI, 2019-2022: Consultancy - Instrumenting Chilean Northern Volcanoes	£80k
Total Grant Capture: of which £354.7k as PI	£893.7k

Supervision:

PhD Supervision – Alyssa Heggison , <i>University of Sheffield</i>	2024-
PhD Supervision – Rebecca England , <i>University of Sheffield</i>	2023-
PhD Supervision – Hannah Calleja , <i>University of Edinburgh</i>	2022-
Supervision & Line Manager – Dr Thomas Wilkes , <i>University of Sheffield</i>	2016-2023
Postdoc Supervision – Dr Tehnuka Ilanko , <i>University of Sheffield</i>	2018-2020

Conference Organisation:

Co-I, Japanese Volcano Monitoring Symposium (150 people), Niigata, Japan 10/2025

Invited Talks:

Pering, T.D., 2021. Low-cost science measuring and studying basaltic volcanism. Lancaster.

Pering, T.D., 2020. Multiparametric measurements of the lava lake at Masaya volcano. EGU.

Impact and Outreach:

REF ICS, Transformative Gas Sensing Technology 2021

Deadly Disasters TV Documentary, Season 2 Ep. 6, [link](#) 2020

Geography Association public lecture “Volcanoes: from inception to eruption”, [link](#) 2020

Academic and Professional Activities

Chair of NERC FSF Steering Committee (member since Sep 2021) 12/2023-

Secretary of Diamond Access Journal Volcanica 04/2026

Editorial Board member at Volcanica 09/2024-

Topic Editor at diamond open access journal Volcanica 12/2020-

Grant and Journal Reviewing

Reviewer for NERC ECR Fellowships 2022-

Reviewer for Chilean Government Grants 2020-

Journal reviewing: Earth and Planetary Science Letters, Journal of Geophysical Research, Journal of Volcanology and Geothermal Hazards, Frontiers in Earth Science, Bulletin of Volcanology.

Teaching and Module Convening:

Statistics: I teach and lead all statistics teaching at L1 and L2 for Geography, I currently have a book contract with SAGE based on the innovation and style of teaching (see Pering and Temple, 2023).

Volcanology: I teach my specialist subject in Applied Volcanology, which incorporates virtual reality.

Methods and Field Teaching: intro geography and field classes in Spain, New Zealand, and the UK.

Selected Administrative Roles:

Director of Undergraduate Studies, *Dept. of Geog., Sheffield* 2023-

UG Admissions Tutor, *Dept. of Geog., Sheffield* 2021-2022

Landscape Dynamics Research Group Lead, *Dept. of Geog., Sheffield* 2019-2023

Teaching Grants:

Alumni Teaching Grant, *embedding virtual reality in the curriculum* £13.6k

Publications: Total = 40, First = 14, Citations = 948, h-index = 19

Key Publications since 2020

Pering, T.D., et al. 2024. The PiGas: A Low-Cost Approach to Volcanic Gas Sampling. Journal of Volcanology and Geothermal Research 449, 108063

Wilkes, T.C., **Pering, T.D.**, et al., 2023. A new permanent, low-cost, low-power SO₂ camera for continuous measurement of volcanic emissions. Frontiers in Earth Sciences 11, 1088992

Calleja, H., **Pering, T.D.**, 2023. Crystals and inclined conduits: analogue experiments for slug-driven volcanism. Volcanica 6 (1), 147-160

- Layana, S., Aguilera, F., Inostroza, M., Tassi, F., Wilkes, T.C., Bredemeyer, S., González, C., **Pering, T.D.**, et al., 2023. Evolution of the magmatic-hydrothermal system at Lastarria volcano (Northern Chile) between 2006 and 2019: Insights from fluid geochemistry. *Frontiers in Earth Science* 11, 1114001
- Wilkes, T.C, **Pering, T.D.**, McGonigle, A.J.S., 2022. Semantic segmentation of explosive volcanic plumes through deep learning. *Computers and Geosciences*, 168 (105216)
- Mason, E., et al., **Pering T.D.**, et al., 2021. Volatile metal emissions from volcanic degassing and lava-seawater interactions at Kilauea Volcano, Hawai'i. *Nature Communications Earth & Environment* 2 (1), 1-16
- Pering, T.D.**, et al., 2020. Combined ground and aerial measurements resolve vent-specific gas fluxes from a multi-vent volcanic system. *Nature Comms*, 11 (1)
- Ilanko, T., **Pering, T.D.**, et al., 2020. Ultraviolet camera measurements of passive and explosive (Strombolian) sulphur dioxide emissions at Yasur volcano, Vanuatu. *Remote Sensing* 12 (17), 2703
- Aguilera, F., et al., **Pering, T.D.**, Ureta, G., 2020. First Measurements of Gas Flux with a Low-Cost Smartphone Sensor-Based UV camera on the Volcanoes of Northern Chile. *Remote Sensing* 12 (13), 2122